

What is Probability?

→ Probability is the measure of chance that an event will happen.

Key Probability Terminology.

1. Experiment

An experiment is an action or process that gives a result.

E.X. selecting 1 student randomly from the dataset.

2. Outcome

Outcome is the result of one trial.

E.X. selected student passed or failed.

3. Sample space

It is a set of all possible outcomes.

E.X. $\{\text{pass, fail}\}$

4. Event

An event is the outcome or group of outcomes that we are interested in.

E.X. student passed the final exam.

Probability event Examples from dataset

- Event 1: student passed the final exam.

A = student passed.

$$P(\text{Pass}) = \frac{\text{No. of passed students}}{\text{Total students}}$$

$$= \frac{130}{200} = 0.65 = 65\%$$

- Event 2: student participated in group discussion.

B = group-discussion = Yes

$$P(\text{group discussion}) = \frac{\text{No. of students with yes}}{\text{Total students}}$$

- Event 3: student has high attendance

C = attendance > 80%.

$$P(\text{High Attendance}) = \frac{\text{No. of students with attendance above 80\%}}{\text{Total students}}$$

Empirical Probability & Theoretical Probability

Formula :

$$\text{Empirical Probability} = \frac{\text{No. of times event occurred}}{\text{Total No. of trials}}$$

If 130 students passed out of 200 students

$$P(\text{Pass}) = \frac{130}{200} = 0.65 = 65\%$$

This is empirical probability because it is based on actual dataset.

Theoretical Probability

If we assume a randomly selected student can either pass or fail with equal chance.

$$S = \{\text{Pass, fail}\}$$

$$P(\text{Pass}) = \frac{1}{2} = 0.5 = 50\%$$

This is theoretical probability because it is based on expected outcomes.

Random variable & Probability distribution

Event : No. of students passing the final exam out of 3 randomly selected students.

Possible values

$$X = 0, 1, 2, 3$$

random variable

because its value depends on random selection of students.

Probability distribution table

- All possible patterns are:

Pattern	Students
FFF	0
PFF	1
FPF	1
FFP	1
PPF	2
PFP	2
FPP	2
PPP	3

$$P(\text{Pass}) = \frac{130}{200} = 0.65$$

$$P(\text{Fail}) = \frac{70}{200} = 0.35$$

For $X=0$, 1 Pattern

$$P(X=0) = 0.35 \times 0.35 \times 0.35 \\ = 0.0429$$

For $X=1$, 3 Pattern

$$P(X=1) = 0.65 \times 0.35 \times 0.35 \\ = 0.0796$$

$$P(X=1) = 3(0.0796) \\ = 0.2388$$

For $X=2$, 3 Pattern

$$P(X=2) = 0.65 \times 0.65 \times 0.35 \\ = 0.1479$$

$$P(X=2) = 3(0.1479) \\ = 0.4436$$

For $X=3$, 1 Pattern

$$P(X=3) = 0.65 \times 0.65 \times 0.65 \\ = 0.2746$$

X	0	1	2	3	
P	0.043	0.239	0.443	0.275	= 1

Mean of random variable

$$E[X] = \sum_{i=1}^n x_i \cdot P(x_i)$$

$$= 0(0.0429) + 1(0.2389) + 2(0.4436) + 3(0.2746)$$

$$= 0 + 0.2389 + 0.8872 + 0.8238$$

$$E[X] = 1.95$$

Variance of random variable

$$\text{Var}(X) = \sum (x_i - \mu)^2 \cdot P(x_i)$$

$\uparrow$
 $E[X]$

$$= 3.8025 \cdot 0.0429 +$$

$$0.9025 \cdot 0.2389 +$$

$$0.0025 \cdot 0.4436 +$$

$$1.1025 \cdot 0.2746$$

$$= 0.1631 + 0.2156 + 0.0011 + 0.3028$$

$$\text{Var}(X) = 0.6826$$

Venn diagram

1. students who study more than 10 hours/ week

$$A \text{ only} = 25$$

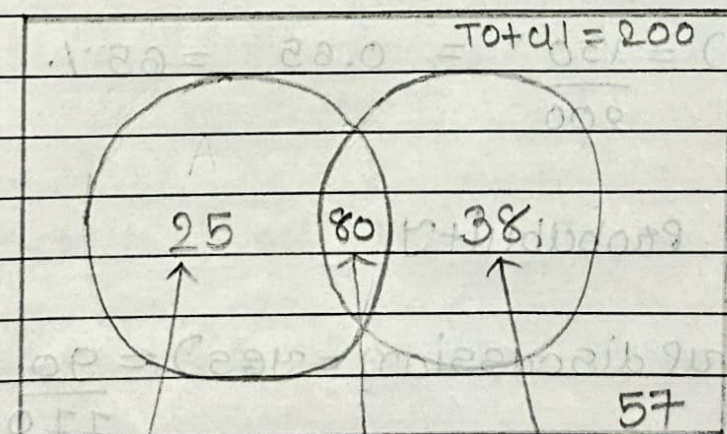
2. students who attend more than 80% of classes

$$B \text{ only} = 38$$

3. overlap showing students who satisfy both conditions

$$A \cap B = 80$$

4. Neither = 57



A only

$A \cap B$

B only

Contingency table

group discussion	Pass	Fail	Total
Yes	90	20	110
No	40	50	90
Total	130	70	200

- Joint Probability

$$P(\text{group discussion} = \text{yes} \cap \text{Pass}) = \frac{90}{200}$$

$$= 0.45$$

$$= 45\%$$

- Marginal Probability

$$P(\text{Pass}) = \frac{130}{200} = 0.65 = 65\%$$

- conditional Probability

$$P(\text{Pass} | \text{group discussion} = \text{yes}) = \frac{90}{110}$$

$$= 0.8181$$

$$= 81.81\%$$

Understanding Relationships

conditional Probability helps us understand the relationship between two events. It tells us how the probability of one event changes when another event is already known.

Q. Identify if participating in group discussion and passing exam are independent, dependent or mutually exclusive events.

→ If 2 events are independent:

$$P(\text{Pass} | \text{group discussion} = \text{yes}) = P(\text{Pass})$$

$$P(\text{Pass} | \text{group discussion} = \text{yes}) = 81.81\%$$

$$P(\text{Pass}) = 0.65 = 65\%$$

since:

$$81.81\% \neq 65\%$$

→ The events are not independent.

→ $\therefore$ These are dependent event.

→ They are not mutually exclusive because a student can both participate in group discussion and pass the exam.

Bayes Theorem Application

suppose:

- 70% of students who pass had high attendance (>80%).

$$P(\text{High attendance} | \text{Pass}) = 0.70$$

- 40% of students who fail also had high attendance.

$$P(\text{High attendance} | \text{Fail}) = 0.40$$

- 60% of all students had high attendance.

$$P(\text{High attendance}) = 0.60$$

Find : $P(\text{Pass} | \text{High attendance})$

$$P(\text{Pass}) = \frac{130}{200} = 0.65$$

$$P(\text{Pass} | \text{High attendance})$$

$$= \frac{P(\text{High attendance} | \text{Pass}) \cdot P(\text{Pass})}{P(\text{High attendance})}$$

$$\text{From } P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

$$= \frac{0.70 \cdot 0.65}{0.60} = \frac{0.455}{0.60}$$

$$= 0.7583 = 75.83\%$$